# Corporate Research Project

AIDAMS - ESSEC & CentraleSupélec - 4th year - 2026-2027

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ENS
ÉCOLE NORMALE SUPÉRIEURE

## Case Description

1. **Title of the case**

Use a Q10 database to develop maps to be used in the land surface model ORCHIDEE

2. **Short description of the case (you can put extensive details in the appendix below). What are the challenges?**

Q10 is a parameter representing the sensitivity of soil organic matter decomposition to temperature. This might be considered as an emerging property of the microbial community but in most of the soil organic decomposition model it is considered as a fixed parameter. Several studies show that this Q10 value actually depends on several factors including environmental factors like climate, soil texture, etc. Nevertheless, in land surface models used within climate models to predict climate change among others, this parameter is fixed in space and time leading to important uncertainties in the model projection. We propose here to use a database of Q10 value measured on different sites and create a machine learning approach to spatialize the values producing then a maps that will be used in the ORCHIDEE land surface model.

3. **Contact of the people in charge of the case (name, email and phone)**

Bertrand Guenet (guenet@geologie.ens.fr)

4. **Contact of the people in charge of the data (name, email and phone)**

Bertrand Guenet (guenet@geologie.ens.fr)

5. **Dataset description (eg list of variables and their definition, total volume of the dataset, source, etc)**

The data will come from a recent publication (Hacopian, M.T., Choreño-Parra, E.M., de Araujo, L.H.A., Zhou, A., Alster, C.J., Randerson, J.T., Treseder, K.K., 2026.
The Q10 of in situ microbial soil respiration varies with mean annual temperature, precipitation, pH, and plant cover: a meta-analysis and spatial prediction of Q10. Sci Rep 16. https://doi.org/10.1038/s41598-026-45615-w)

6. **Other useful information (How frequent can you meet the group of students? Can the group work on site? etc.)**

To be determined but likely once every 2 weeks.

7. **Date, time and place of the Kick-off meeting (give a phone number just in case)**

Date and time will be determined later but the meeting will be at the Geoscience department of Ecole Normale Supérieure, 24 rue Lhomond, 75 005 Paris

8. **How will you measure the success of the case solution?**

Based on statistical metrics such as R2, RMSE, AIC etc.

# Appendix

Please feel free to add more details on the case, contextual articles to read, examples or extracts of data:

# Contacts

Program leads


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